Library Management Project in ServiceNow

Abstract:

The Library Management Project in ServiceNow streamlines the process of borrowing and managing books by integrating role-based access, custom tables, automated workflows, and reporting. Roles such as student and librarian define access levels, while tables like Book and Borrow Request store catalog and transaction data. Automation through Flow Designer ensures efficient approvals, and Dot Walking enhances usability by dynamically retrieving related information. The solution improves transparency, reduces manual effort, and provides administrators with actionable insights.

Objectives:

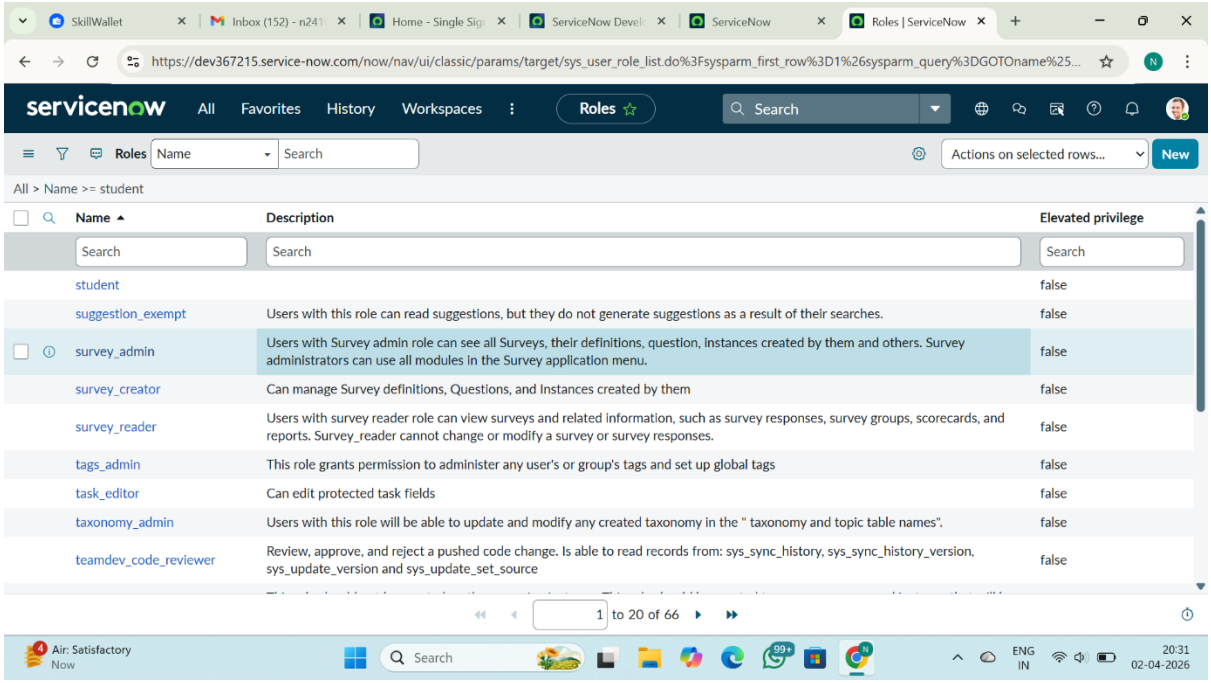
- Define roles (student, librarian) to enforce controlled access.
- Create and configure custom tables for catalog and borrow requests.
- Automate approvals and book issuance using Flow Designer.
- Apply UI Policies and ACLs to maintain consistency and security.
- Provide reporting capabilities for administrators to track library usage.

Technology Description:

- Roles: Student and librarian roles manage access permissions.
- Custom Tables: Book and Borrow Request tables store catalog and transaction data.
- Reference Qualifier: Ensures only available books can be requested.
- Flow Designer: Automates approval workflows and updates book status.
- UI Policies: Lock or enforce field behavior after approval.

- Access Control Rules (ACLs): Enforce role-based permissions for read, write, and delete operations.

- Reports: Generate insights such as Most Borrowed Books.



Detailed Execution of Project Phases:

Phase 1: Role Creation:

- Navigate to User Administration -> Roles.

- Create roles: librarian and student.

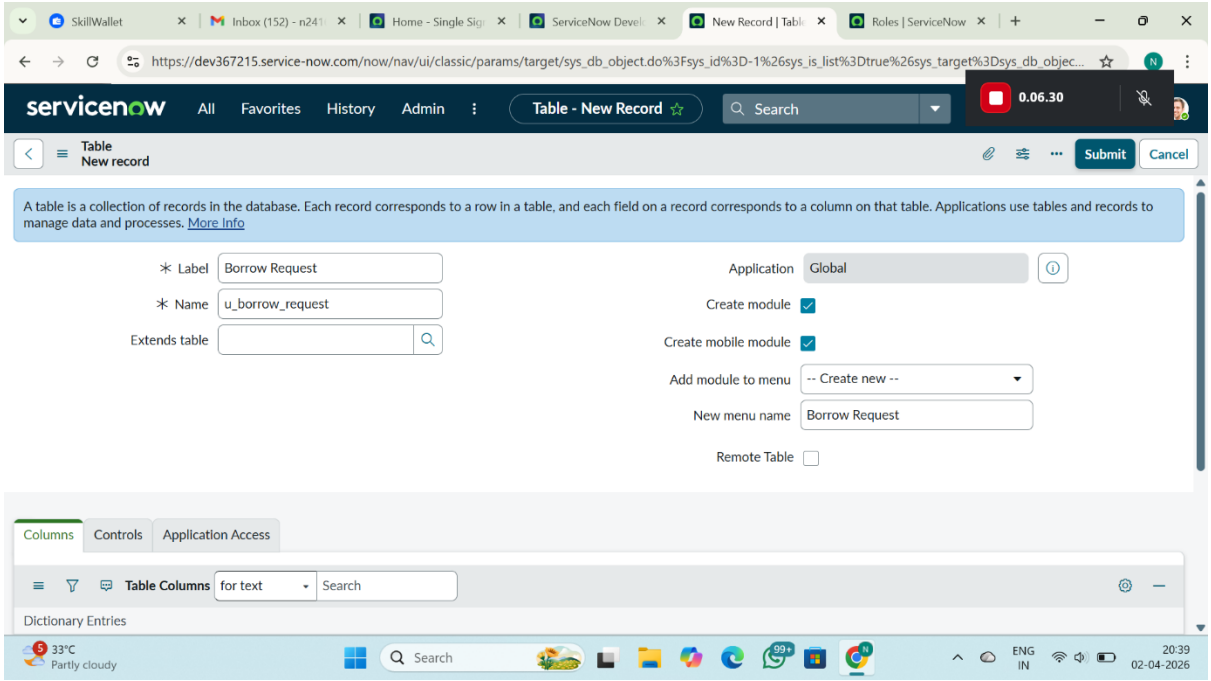
- Assign roles to users as required.

Phase 2: Table Creation:

- Navigate to System Definition -> Tables.

- Create tables: Book and Borrow Request.

- Configure fields such as Title, Author, Status, and Return Date.



Phase 3: Reference Qualifier:

- Configure the Borrow Request table's Book field to reference only records where Status = Available.

Phase 4: Flow Designer Automation:

- Create a flow named Borrow Request Approval Flow.

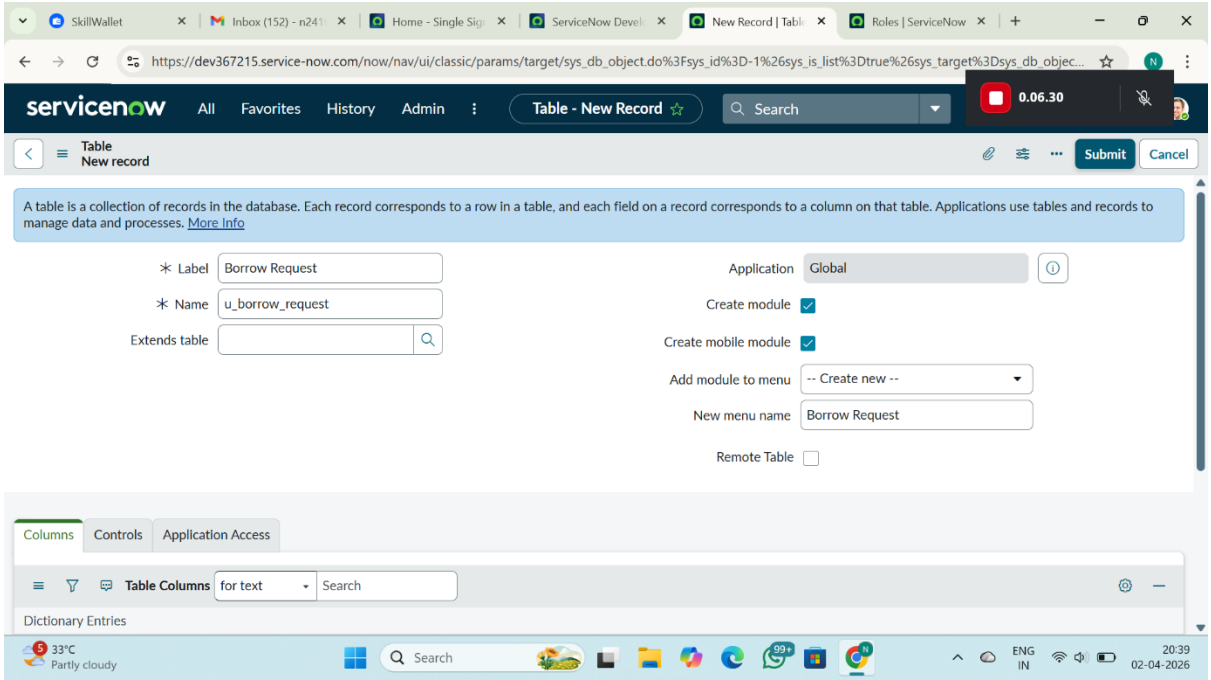
- Trigger: Borrow Request created with Status = Requested.

- Actions.

- Approval by librarian.

- On approval, update Book.Status = Issued and Borrow Request.Status = Approved.

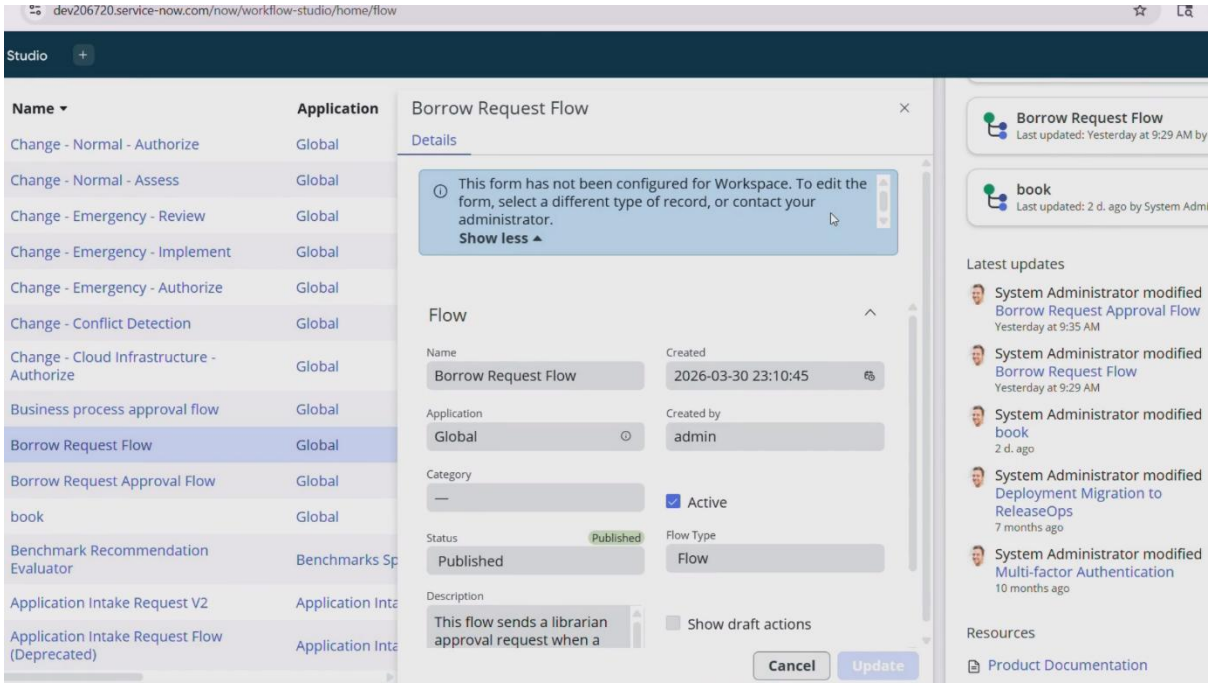
- Send email notification to student.



Phase 5: UI Policies:

- Create a UI Policy for Borrow Request.

- Example: Make Return Date mandatory when Status = Issued.

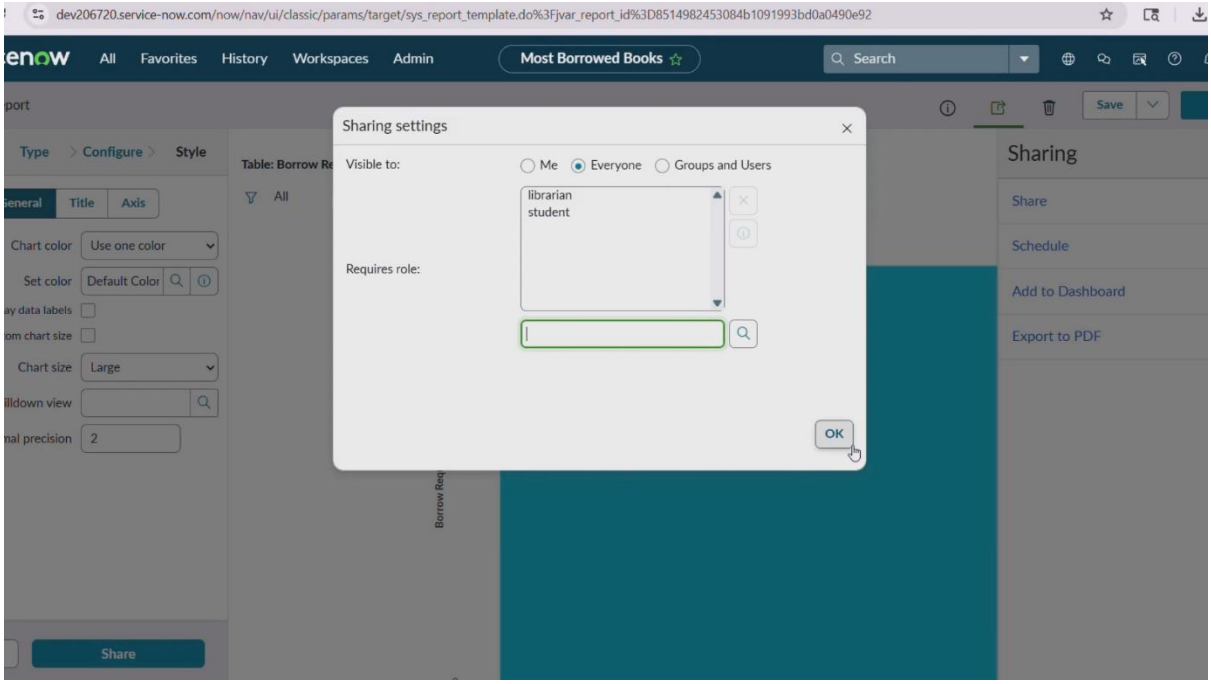


Phase 6: Access Control Rules:

- Configure ACLs for Book table.

 - Students: read only.

- Librarians: read, write, delete.



Phase 7: Reporting:

- Create a report named Most Borrowed Books.

- Source: Borrow Request table.

- Group by Book, aggregate by Count, filter Status = Approved.

Project Explanation:

The project integrates multiple ServiceNow features to build a functional library management system. Roles ensure secure access, tables store structured data, and reference qualifiers maintain integrity. Flow Designer automates approvals and notifications, while UI Policies and

ACLs enforce consistency and security. Reporting provides administrators with valuable insights into library usage patterns.

Conclusion:

The Library Management Project successfully optimizes book borrowing and management processes in ServiceNow. By combining automation, role-based access, and reporting, the system enhances efficiency, accountability, and transparency. Students benefit from a simplified request process, while librarians gain streamlined approval workflows.

Future Scopes:

- Scheduled Jobs: Automate recurring tasks such as overdue reminders.
- Integration with Digital Resources: Extend to e-books and online catalogs.
- Enhanced Notifications: SMS or mobile app alerts for approvals and returns.
- Advanced Analytics: Track borrowing trends, user activity, and resource utilization.
- Scalability: Extend framework to other academic services such as reservations or asset tracking.